

7 Warfarin is used in the treatment of cardiovascular disease.

Warfarin inhibits the synthesis of prothrombin.

(a) Describe the role of prothrombin in the blood clotting process.

(3)

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(b) Which type of treatment is warfarin?

(1)

- ☐ A anticoagulant
- ☐ B antihypertensive
- ☐ C platelet inhibitor
- ☐ D statin

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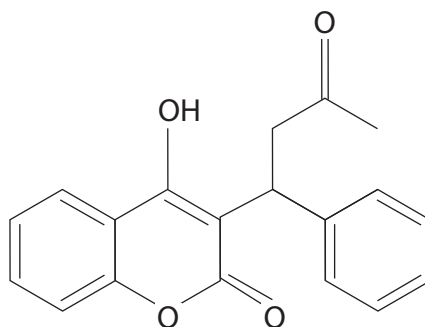
- (c) Reduced vitamin K is needed for the synthesis of prothrombin.

An enzyme, vitamin K epoxide reductase (VKOR), converts vitamin K to reduced vitamin K.

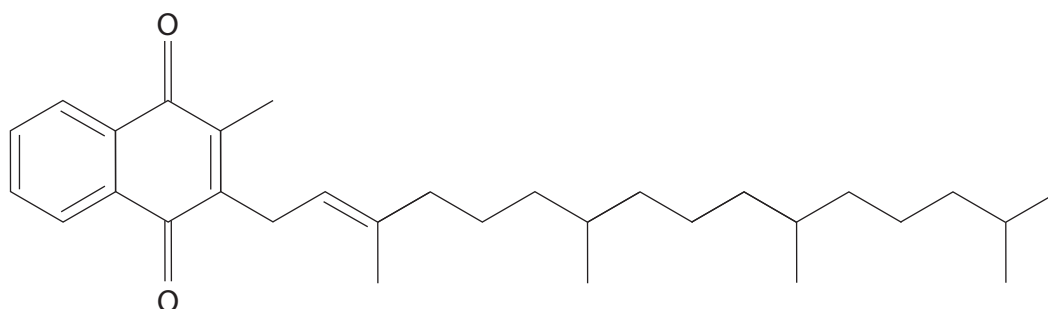
The diagram shows this conversion.



- (i) The diagrams show the structures of warfarin and vitamin K.



Warfarin



Vitamin K

Using the information in the diagrams, suggest why warfarin inhibits the synthesis of prothrombin.

(2)

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- (ii) Suggest why people taking warfarin have to avoid eating foods that contain a high concentration of vitamin K.

(2)

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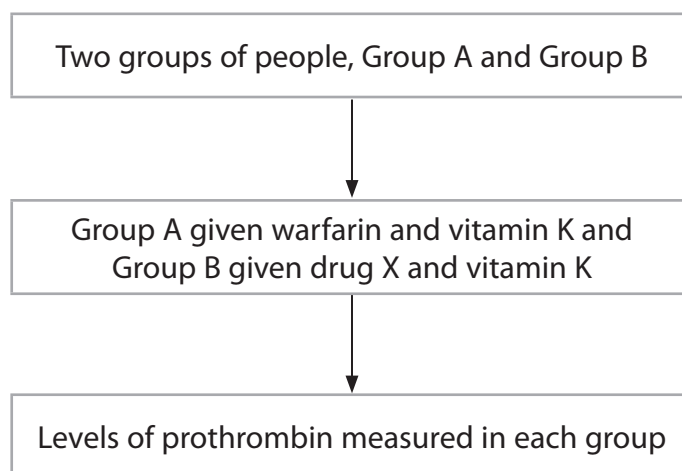
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(d) Other drugs are available that work in a similar way to warfarin.

The diagram shows one study to compare the effect of warfarin with drug X.



Explain how this study should be designed so that the effectiveness of these two drugs can be compared.

(4)

(Total for Question 7 = 12 marks)

